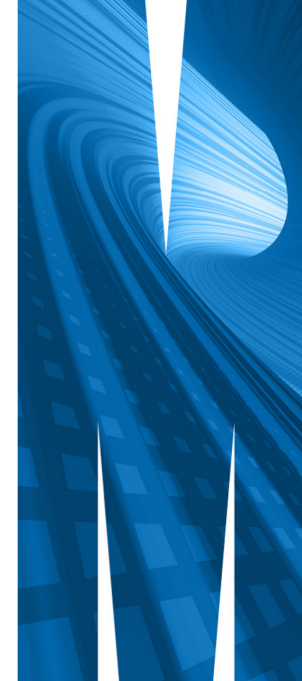
MONASH University

ETX1100/ETX5900 Business Statistics

Week 7: From regression to business decisions

19 January 2027



Outline

1. Business question and evidence
2. Estimate demand with the Analysis ToolPak
3. Formulate the business model
4. Compare and challenge decisions
5. Integrated classroom exercise
6. Consolidation

Today’s journey

1. Use the Analysis ToolPak to estimate a price-demand relationship.
2. Turn regression output into a transparent spreadsheet model.
3. Compare listed prices subject to production capacity.
4. Test how the recommendation changes when assumptions change.
5. Communicate a decision and its limits.

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Retrieval check

- Which variable is the outcome in a demand regression?
- What do the intercept and price coefficient represent?
- Why should a prediction stay within the observed price range?
- What is the difference between a fitted value and an actual sale?

The pricing question

A business wants to choose a listed price for the next period.

It has a synthetic pricing pilot with 24 comparable store-weeks. Different prices were assigned across those store-weeks; units sold were recorded.

The business also knows its variable cost, fixed cost, and weekly production capacity. Recommend a **feasible price from the listed options** and explain how uncertain demand affects the recommendation.

The workflow

Business question $\rightarrow$ pilot data $\rightarrow$ ToolPak regression $\rightarrow$ spreadsheet formulas $\rightarrow$ feasible scenario comparison $\rightarrow$ sensitivity $\rightarrow$ recommendation

Regression supplies a demand estimate. The spreadsheet translates that estimate into revenue, cost, and profit. Neither part replaces business judgement.

Watch me: inspect the pilot

Open the Week 7 workbook, sheet **Pilot sales**.

1. Identify the observational unit, response, and predictor.
2. Check units, missing values, and the observed price range.
3. Plot units sold against price.
4. Describe direction, form, spread, and unusual points.

Now you try: data and decision audit

Time allowed: 6 minutes

Record the pilot’s price range, one reason a straight-line model may be useful, one reason its prediction may be wrong, and one business input the pilot does not supply.

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Watch me: run Regression

In Excel choose **Data > Data Analysis > Regression**.

- **Input Y Range:** units sold, including its label.
- **Input X Range:** price, including its label.
- Check **Labels** and select a blank output area or new worksheet.
- Ask for residual output if available.

Keep the raw pilot data unchanged.

What the output gives us

Write the fitted model as

$$\widehat{Units} = b_0 + b_1 \times Price.$$

Read b_0 and b_1 from the **Coefficients** column. Check R^2, the price coefficient's p-value, and whether residuals show an obvious pattern.

The intercept may have no practical meaning at price zero. It is still needed in the prediction formula.

Now you try: explain the regression

Time allowed: 8 minutes

Run the ToolPak regression and report the fitted equation with units, the expected change in units for an AUD 1 price increase, a predicted quantity at an observed price, one output check, and one limit on treating the estimate as future demand.

Watch me: validate before using coefficients

- Confirm that **Y** is units and **X** is price.
- Compare a fitted value with the corresponding observed quantity.
- Scan residuals for structure and outliers.
- Check predictions in the proposed price range are non-negative.
- Keep enough coefficient precision to avoid changing close decisions by rounding.

Now you try: output quality check

Time allowed: 6 minutes

Choose two pilot observations. Calculate fitted units and residuals from your coefficients. Compare them with Excel’s regression output. Flag anything that does not agree.

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Model anatomy

Component	In this case
Evidence	pilot price and sales
Inputs	coefficients, costs, capacity
Decision	listed selling price
Calculations	units, revenue, cost, profit
Objective	highest listed feasible profit
Rules	whole non-negative units, capacity, pilot range

Watch me: connect output to inputs

Open **Decision model**. Paste the ToolPak intercept into cell B7 and price slope into B8. The business assumptions are in B4:B6.

For a listed price in A13, enter the predicted units formula in B13:

```
=$B$7+$B$8*A13
```

Check one prediction by hand. Fill down only after the first row is correct.

Now you try: label and audit the model

Time allowed: 6 minutes

Identify which inputs came from data and which are business assumptions. Verify the units; test one low and one high listed price; explain why a prediction outside the pilot range needs extra caution.

Watch me: apply the capacity rule

Stock is counted in whole units and cannot exceed capacity. Enter in C13:

```
=MAX(0,MIN(ROUNDDOWN(B13,0),$B$6))
```

This rounds predicted demand down, caps units at the capacity in B6, and prevents negative planned sales. Check that this is the intended business rule.

Watch me: calculate profit

- Revenue = price times units planned.
- Cost = fixed cost plus unit cost times units planned.
- Profit = revenue minus cost.

In the first scenario row, use `=A13*C13` in D13, `=$B$5+$B$4*C13` in E13, and `=D13-E13` in F13. Check the first row before copying down.

Now you try: complete the scenario model

Time allowed: 9 minutes

Fill columns B:F for every listed price. Check that units planned are whole, non-negative, never exceed capacity, and that profit agrees with an independent calculation. In column G, flag negative predicted demand or a capacity breach. Name one formula that could be valid in Excel but wrong for the business.

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Watch me: find the best listed scenario

Filter or sort the completed rows by predicted profit. Keep the check column visible.

Compare the highest-profit listed price with its neighbouring listed prices. Report price, predicted units, units planned, expected profit, and whether capacity binds.

This is a **comparison of listed options**, not a claim that every possible price has been searched.

Now you try: defend a recommendation

Time allowed: 8 minutes

Recommend one listed price. Report the nearest alternative and its profit difference, whether capacity changes the sales forecast, and one assumption that could reverse the ranking.

Why sensitivity matters

The regression estimates demand from limited data. Costs and capacity may also change.

Change **one input at a time**, recalculate the scenario formulas, and record the best listed feasible option. Restore baseline inputs after each trial.

Watch me: stress-test one assumption

Increase unit variable cost in B4 from AUD 40 to AUD 50.

Before changing the cell, predict whether the recommended price is likely to rise, fall, or stay the same. Then recalculate and compare the best listed price and profit with baseline.

Do not overwrite the ToolPak coefficients while testing a cost assumption.

Now you try: two stress tests

Time allowed: 10 minutes

Starting from baseline each time:

1. Set capacity to 200 units.
2. Increase unit variable cost to AUD 50.

For each case, report the best listed price, planned units, and profit. Explain which assumption changes the decision more and why.

A model is a conditional recommendation

The chosen price depends on the pilot reflecting future customers; prices being comparable across store-weeks; a roughly linear relationship within the observed range; and costs and capacity being accurate.

A small p-value or high R^2 cannot settle these business assumptions.

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Group brief: recommend a launch price

Prepare a one-page pricing brief for the manager using your workbook. Include:

1. the ToolPak equation and one output check;
2. a labelled formula model and best listed feasible price;
3. one baseline alternative and both sensitivity results; and
4. a recommendation, one limitation, and a revisit condition.

Now you try: team working time

Time allowed: 15 minutes

Complete the brief. Each group member should be able to explain where the regression coefficients came from and how the workbook turns them into a decision.

Watch me: critique a decision memo

Check whether a memo gives a listed feasible price and units, distinguishes predicted from actual profit, states the capacity rule, compares sensitivity against the same baseline, and avoids unsupported causal or global-optimum claims.

Now you try: improve the memo

Time allowed: 7 minutes

Swap briefs with another group. Identify one calculation to verify, one claim to qualify, and one action that would make the recommendation more useful. Revise your final paragraph.

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Common traps

- Reversing the ToolPak X and Y ranges.
- Copying a rounded coefficient that changes close rankings.
- Applying the regression far outside pilot prices.
- Hard-coding assumptions inside every scenario formula.
- Forgetting fixed cost or capacity.
- Calling the best listed option an unrestricted optimum.
- Treating a prediction or statistical association as guaranteed sales.

Closing check

Can you obtain and check a ToolPak regression output; identify inputs, a decision, calculations, an objective, and rules; build and audit a formula-based spreadsheet model; select the best listed feasible scenario; test assumptions; and communicate a recommendation with limitations?